



Lavanya R 250701385-2025-CSE ▾

L2**Started on** Tuesday, 11 August 2026, 12:30 PM**State** Finished**Completed on** Friday, 21 August 2026, 8:19 AM**Time taken** 9 days 19 hours**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00**Problem Statement**

Given an array of 1s and 0s this has all 1s first followed by all 0s. Aim is to find the number of 0s. Write a program using Divide and Conquer to Count the number of zeroes in the given array.

Input Format

First Line Contains Integer m – Size of array

Next m lines Contains m numbers – Elements of an array

Output Format

First Line Contains Integer – Number of zeroes present in the given array.

Answer: (penalty regime: 0 %)

```

1  #include<stdio.h>
2  int main(){
3      int n;
4      scanf("%d",&n);
5      int arr[n];
6      for(int i=0;i<n;i++){
7          scanf("%d",&arr[i]);
8      }
9      int c=0;
10     for(int i=0;i<n;i++){
11         if(arr[i]==0){
12             c++;
13         }
14     }
15     printf("%d",c);
16 }
```

	Input	Expected	Got	
✓	5 1 1 1 0 0	2	2	✓
✓	10 1 1 1 1 1 1 1 1 1 1 1	0	0	✓

	Input	Expected	Got	
✓	8 0 0 0 0 0 0 0 0 0	8	8	✓
✓	17 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0	2	2	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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Started on	Tuesday, 11 August 2026, 12:41 PM
State	Finished
Completed on	Tuesday, 11 August 2026, 12:44 PM
Time taken	3 mins 5 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array `nums` of size `n`, return *the majority element*.

The majority element is the element that appears more than $\lfloor n / 2 \rfloor$ times. You may assume that the majority element always exists in the array.

Example 1:

Input: `nums = [3,2,3]`

Output: 3

Example 2:

Input: `nums = [2,2,1,1,1,2,2]`

Output: 2

Constraints:

- `n == nums.length`
- `1 <= n <= 5 * 104`
- `-231 <= nums[i] <= 231 - 1`

For example:

Input	Result
3 3 2 3	3
7 2 2 1 1 1 2 2	2

Answer: (penalty regime: 0 %)

```

1  #include<stdio.h>
2  int main(){
3      int n,a[100],candidate,count=0;
4      scanf("%d",&n);
5      for(int i=0;i<n;i++)
6          scanf("%d",&a[i]);
7      for(int i=0;i<n;i++){
8          if(count==0){
9              candidate=a[i];
10             count=1;
11         }
12         else if(a[i]==candidate)
13             count++;
14         else
15             count--;
16     }
17     printf("%d",candidate);
18     return 0;
19 }
```

	Input	Expected	Got	
✓	3	3	3	✓
	3 2 3			

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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L2

Started on	Tuesday, 11 August 2026, 12:45 PM
State	Finished
Completed on	Friday, 21 August 2026, 8:22 AM
Time taken	9 days 19 hours
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00**Problem Statement:**

Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Value for x

Output Format

First Line Contains Integer – Floor value for x

Answer: (penalty regime: 0 %)

```

1  #include<stdio.h>
2  int main(){
3      int n;
4      scanf("%d",&n);
5      int arr[n];
6      for(int i=0;i<n;i++){
7          scanf("%d",&arr[i]);
8      }
9      int x;
10     scanf("%d",&x);
11     for(int i=0;i<n;i++){
12         if(arr[i]>x){
13             printf("%d",arr[i-1]);
14             break;
15         }
16     }
17 }
```

	Input	Expected	Got	
✓	6 1 2 8 10 12 19 5	2	2	✓
✓	5 10 22 85 108 129 100	85	85	✓

	Input	Expected	Got	
✓	7	9	9	✓
	3			
	5			
	7			
	9			
	11			
	13			
	15			
	10			

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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L2**Started on** Tuesday, 11 August 2026, 6:24 PM**State** Finished**Completed on** Tuesday, 11 August 2026, 7:00 PM**Time taken** 35 mins 38 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00**Problem Statement:**

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as "No".

Note: Write a Divide and Conquer Solution

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

Output Format

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value "x")

Answer: (penalty regime: 0 %)

```

1  #include<stdio.h>
2  int bs(int a[],int l,int h,int x,int ex)
3  {
4      if(l>h)
5          return -1;
6      int m=(l+h)/2;
7      if(a[m]==x&&m!=ex)
8          return m;
9      if(a[m]<x)
10         return bs(a,m+1,h,x,ex);
11         return bs(a,l,m-1,x,ex);
12     }
13 int main(){
14     int n,x,i,j;
15     scanf("%d",&n);
16     int a[n];
17     for(i=0;i<n;i++)
18         scanf("%d",&a[i]);
19     scanf("%d",&x);
20     for(i=0;i<n;i++){
21         j=bs(a,0,n-1,x-a[i],i);
22         if(j!=-1){
23             printf("%d\n%d",a[i],a[j]);
24             return 0;
25         }
26     }
27     printf("No");
28 }
```

	Input	Expected	Got	
✓	4 2 4 8 10 14	4 10	4 10	✓
✓	5 2 4 6 8 10 100	No	No	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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L2

Started on	Tuesday, 11 August 2026, 7:00 PM
State	Finished
Completed on	Friday, 21 August 2026, 8:29 AM
Time taken	9 days 13 hours
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Write a Program to Implement the Quick Sort Algorithm

Input Format:

The first line contains the no of elements in the list-n

The next n lines contain the elements.

Output:

Sorted list of elements

For example:

Input	Result
5 67 34 12 98 78	12 34 67 78 98

Answer:

```

1  #include<stdio.h>
2  int main(){
3      int n;
4      scanf("%d",&n);
5      int arr[n];
6      for(int i=0;i<n;i++){
7          scanf("%d",&arr[i]);
8      }
9      int t;
10     for(int i=0;i<n;i++){
11         for(int j=0;j<n;j++){
12             if(arr[i]<arr[j]){
13                 t=arr[i];
14                 arr[i]=arr[j];
15                 arr[j]=t;
16             }
17         }
18     }
19 }
20 for(int i=0;i<n;i++){
21     printf("%d ",arr[i]);
22 }
23 }
24

```

	Input	Expected	Got	
✓	5 67 34 12 98 78	12 34 67 78 98	12 34 67 78 98	✓
✓	10 1 56 78 90 32 56 11 10 90 114	1 10 11 32 56 56 78 90 90 114	1 10 11 32 56 56 78 90 90 114	✓
✓	12 9 8 7 6 5 4 3 2 1 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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